Evaluating the Effectiveness of an Automated Cell Counting Program

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Background:

- Neuroscience research relies heavily on cell counting to assess brain injury and evaluate neuroprotective treatments.
- Manual methods (i.e. hand counting) have been used traditionally, but automated programs offer the potential for standardized and error-free data analysis.
- In the context of studying hypoxic-ischemic encephalopathy (HIE), a prevalent brain injury in infants, we aimed to assess the accuracy of an automated cell counting program in an *in vitro* slice culture ferret model of brain injury and treatment.

Statistical Tests:

- Bland-Altman and linear regression tests were used.
- The overall accuracy of the macro was compared to historic hand counted data for DAPI stained cells (or total cell count).
- Term slices were single channel images. Preterm slices were multicolor images.

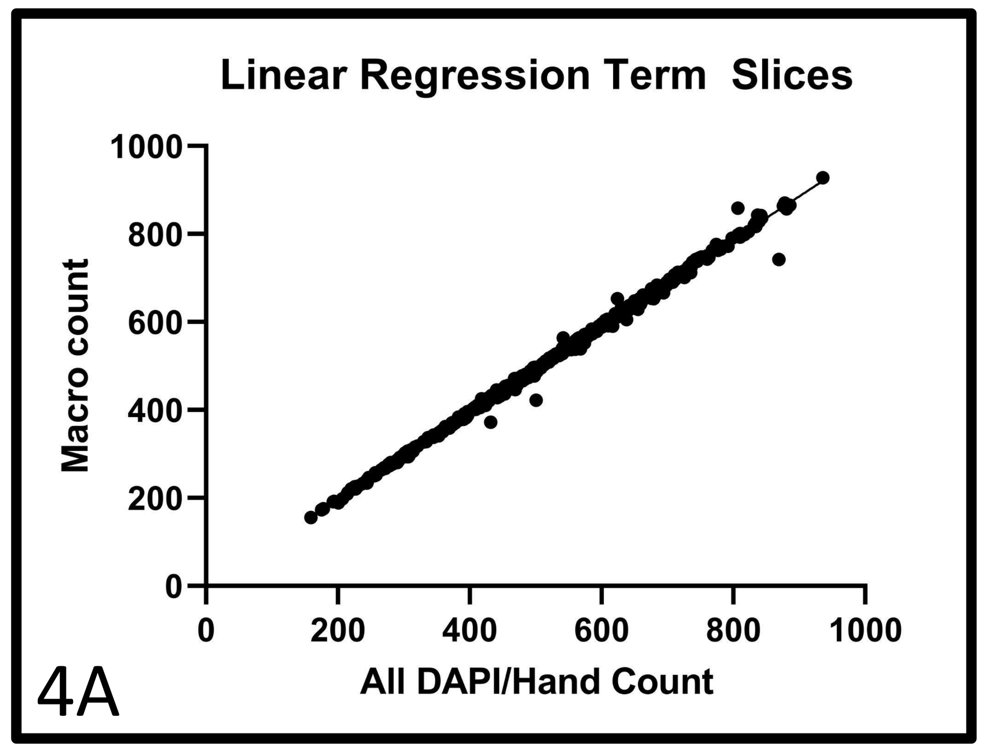


Figure 4A: Macro cell counts and hand counts showed excellent agreement (R^2=0.9965). Term slices are from P21 ferrets administered caffeine as treatment.

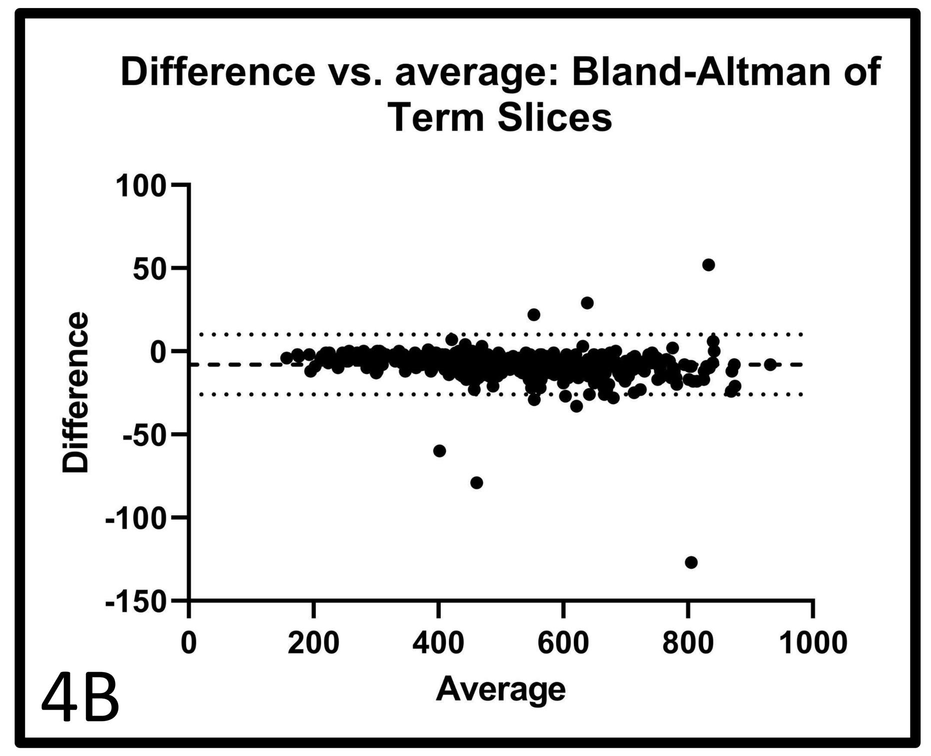


Figure 4B: Macro cell counts and hand counts showed a small average difference (Avg. diff. -7.969, 95% CI [-25.96-10.02]). Term slices are from P21 ferrets administered caffeine as treatment.

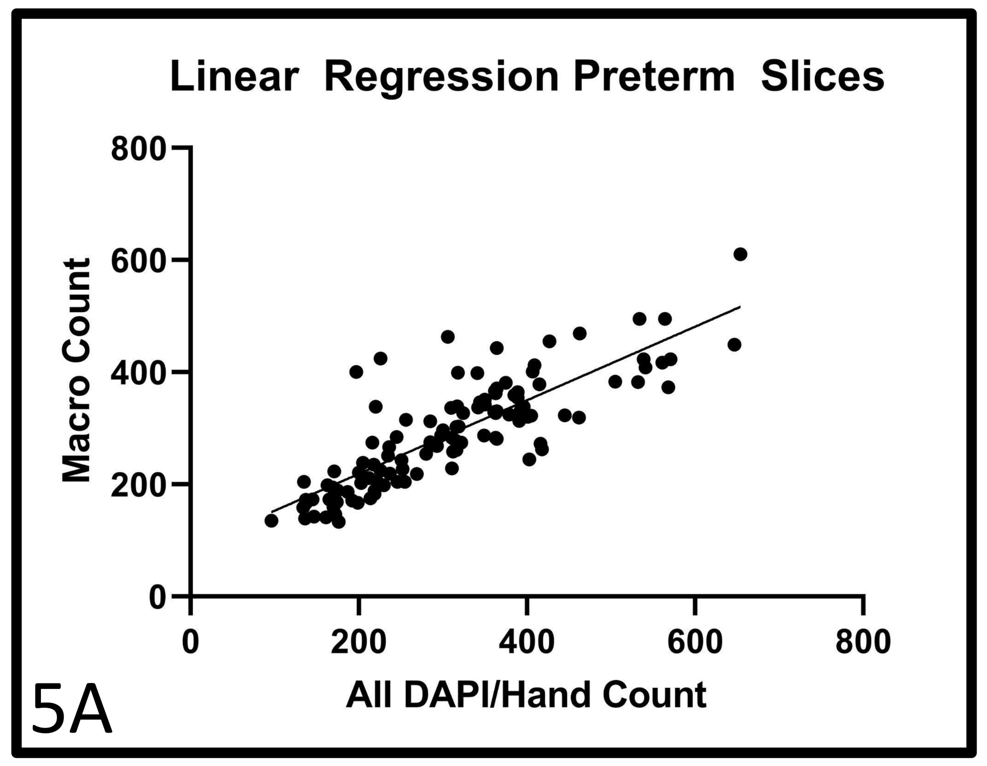


Figure 5A: Macro cell counts and hand counts showed good agreement (R^2=0.6904). Preterm slices are from P14 ferrets administered azithromycin as treatment.

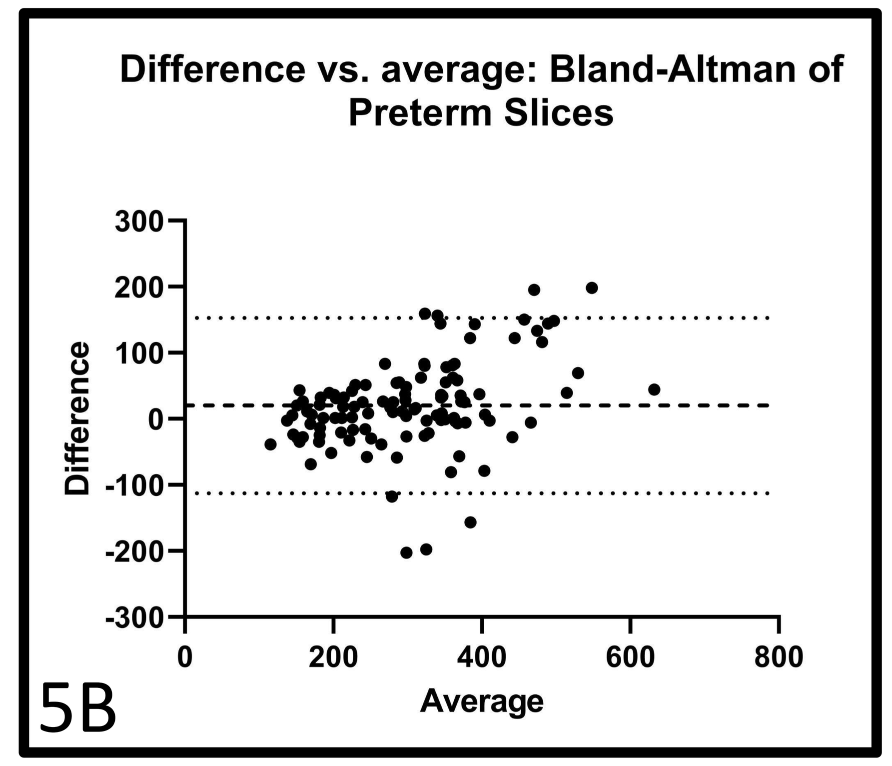


Figure 5B: Macro cell counts and hand counts showed a small average difference with wider CI than preterm data. (Avg. diff. 20.00, 95% CI [-112.6-152.6]). Preterm slices are from P14 ferrets administered azithromycin as treatment.

Methods: Figure 1

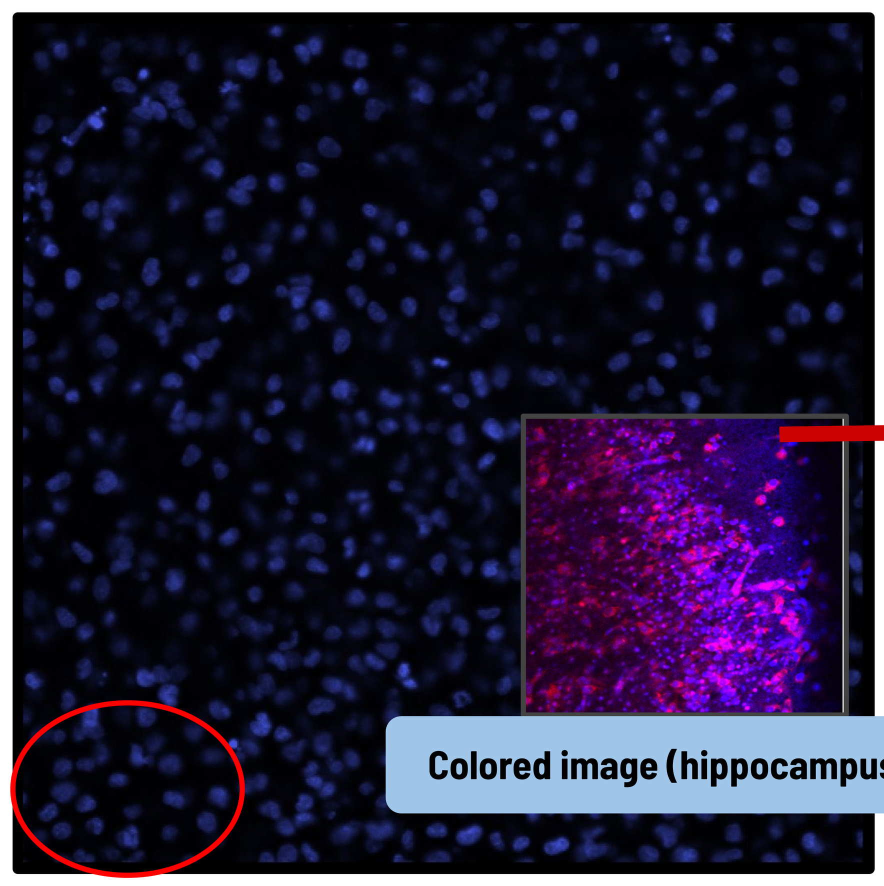
The program runs locally as a macro in the free software ImageJ/Fiji and searches through folders/subfolders for images.

Step 1: Figure 2

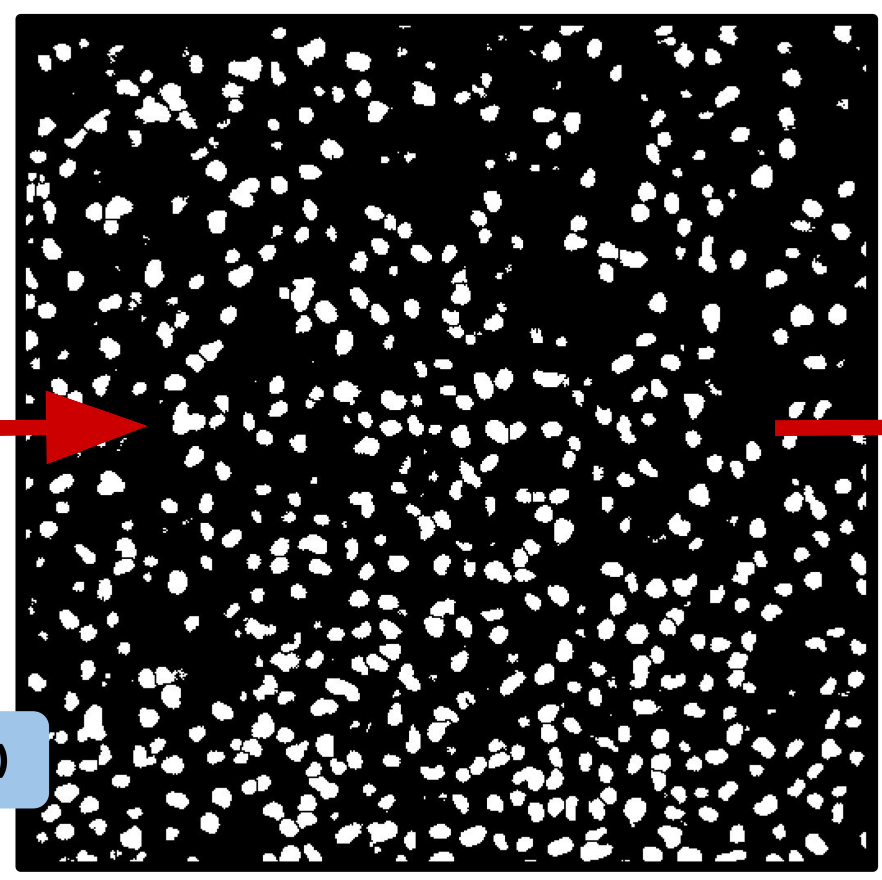
Converts images into a binary image based on fluorescence threshold data (used to stain cell nuclei) and applies the ImageJ function “watershed” that breaks large groups into smaller groups.

Step 2: Figure 3

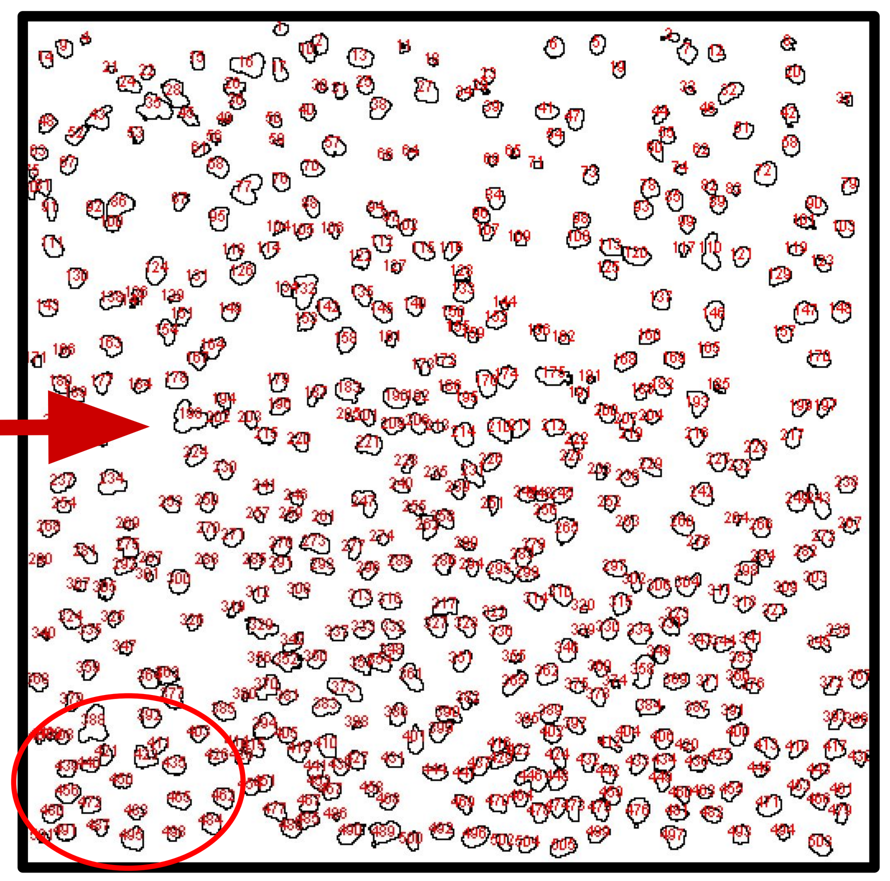
Runs the “analyze particles” function which outputs a total cell count based on the size and circularity of the cells, and then saves the final image.



Original image (basal ganglia)



- Fluorescence threshold, cellular size, and circularity values were determined before data collection then kept consistent between studies. Otsu auto-thresholding was used for multicolor images.



Output image (Final count: 505)

Results:

- Linear regression results (Fig. 4A) indicate the program is extremely accurate and precise in single channel term slices and less precise in multicolor preterm slices (Fig. 5A).
- Bland-Altman statistical tests indicate the average difference between counts was near 0 in both groups (Fig. 4B, Fig 5B).

Future Steps/Discussion:

- Validating this automated method represents a significant advancement in research methodology as cell counting programs can save labs time and resources.
- Results suggest that the macro works best on images with a clear, non-pixelated blue DAPI channel. More testing will determine which types of images work best.
- Current limitations in our research involve differentiating between healthy and dead or dying cells, which would be an important future step for automated cell counting.

QR Code with Video of Program Running and link to code

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Acknowledgements:
Firstly I would like to thank my wonderful teammates at the NNL lab and especially my incredible PI Dr. Wood. This project would have never been possible without their support and guidance. Special thanks to the Nance lab and to all the researchers who contributed to the cell count data I used for my project. ChatGPT and online forums helped with troubleshooting. Another well deserved thanks to Olivia B. who helped me troubleshoot and run my statistical tests.

- Code templates from ImageJ/Fiji were taken and modified using ChatGPT, and other snippets were used and modified from online forums like GitHub.